4190.428 Introduction to Machine Learning

# Feature Representation

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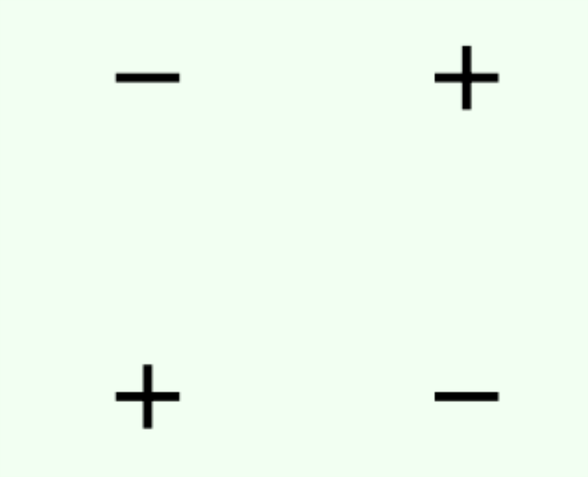
# Overview

- Linear classifiers
  - Easy to work with and analyze
  - Very restricted class of hypothesis
- Can we consider a better approach?

# XOR dataset

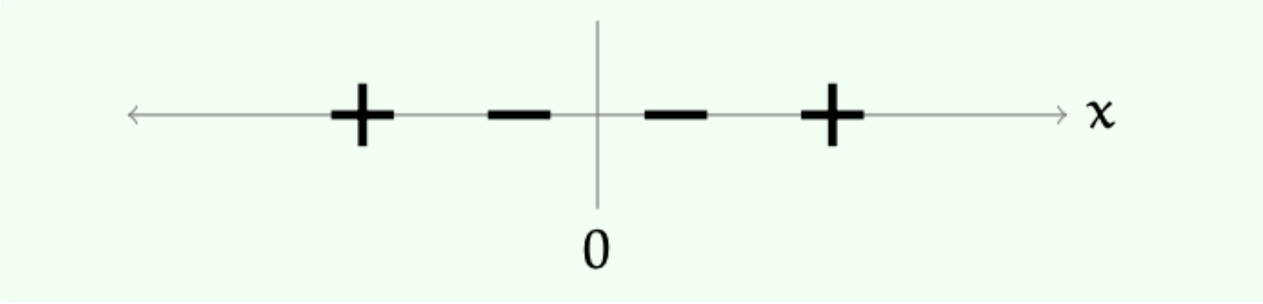
- Let’s starts with exclusive or (XOR) dataset




- There is no linear separator for this two-dimensional dataset!

# XOR dataset

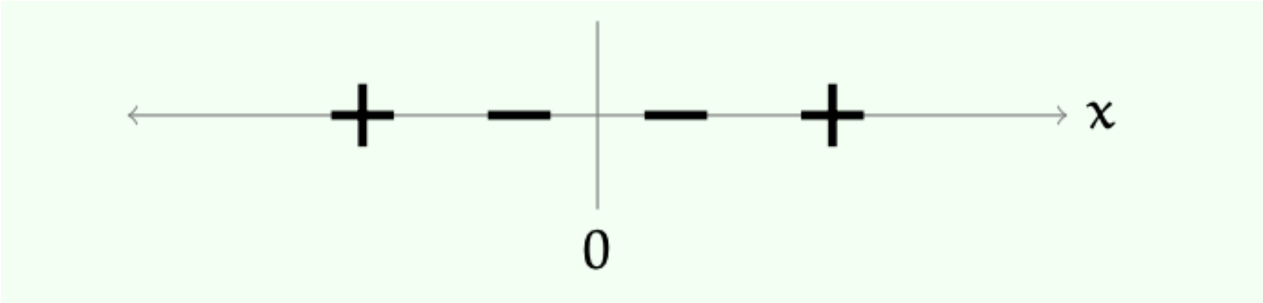
- Example) Exclusive or (XOR) dataset
  - Trick: take a low-dimensional dataset and move it, using a non-linear transformation into higher-dimensional space
  - Let's assume we have now a new dataset starts in 1-D




- Obviously, it is still not linearly separable

# XOR dataset

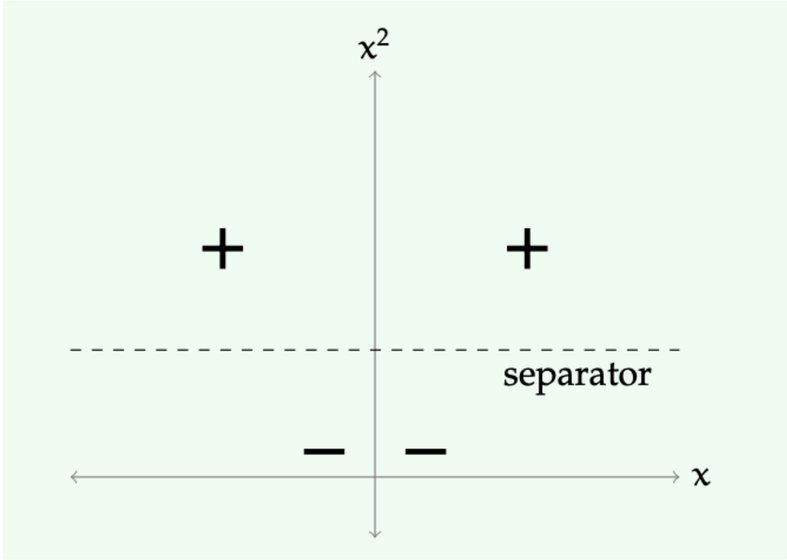
- Example) Exclusive or (XOR) dataset
  - Let's assume we have now a new dataset starts in 1-D




- Idea: Consider the transformation $\phi(x) = [x, x^2]$

# Transformed XOR dataset

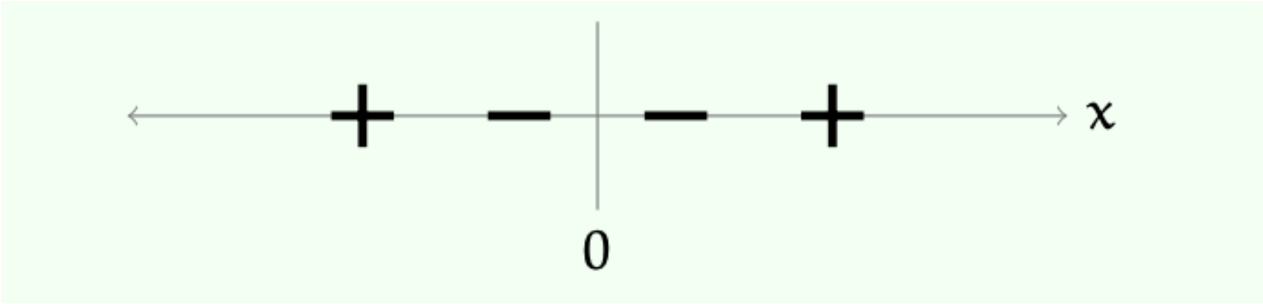
- Let's consider the transformation $\phi(x) = [x, x^2]$
  - Now separable with linear separator (lots of more)
  - We are now in $\phi$ space!




- **A linear separator** in $\phi$ space is **a nonlinear separator** in the original space

# Separator in the original space

- Consider separator $x^2 - 1 = 0$
  - Which labels $x^2 - 1 > 0$ as positive label negative otherwise
- What separator does it corresponds to in the original 1D space?



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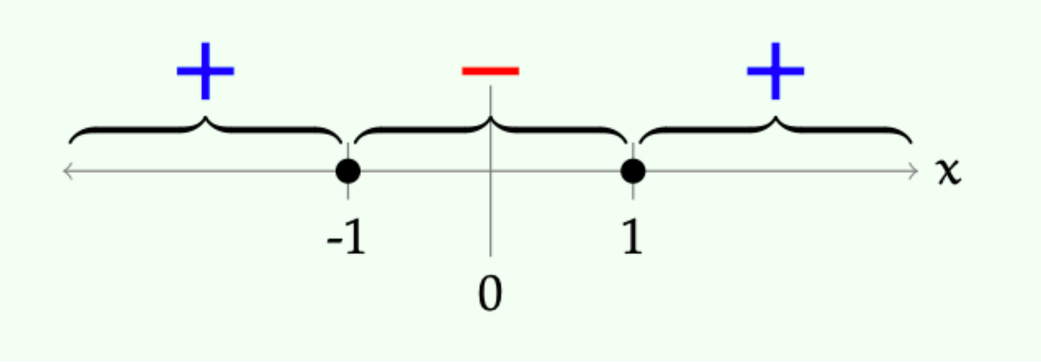



- It is $-1$ and $+1$. Those two points constitute our separator, back in the original space.

# Kernel Methods

- This strategy is very general and widely useful!
- A basis for **Kernel methods**
- A motivation for **multi-layer neural networks**
- Question: Is other ways to construct $\phi$?

# Polynomial Basis

- If the features in your problem are already naturally numerical
- One systematic strategy is **polynomial basis**
- **K-th order basis** (where k is positive) includes a feature for every possible product of k different dimensions

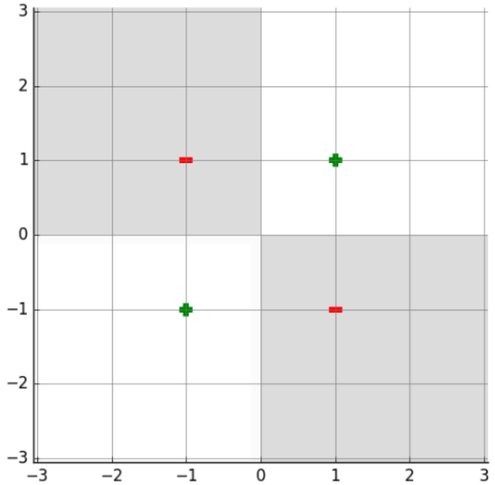
| Order | $d = 1$ | in general |
|---|---|---|
| 0 | $[1]$ | $[1]$ |
| 1 | $[1, x]$ | $[1, x_1, \ldots, x_d]$ |
| 2 | $[1, x, x^2]$ | $[1, x_1, \ldots, x_d, x_1^2, x_1x_2, \ldots]$ |
| 3 | $[1, x, x^2, x^3]$ | $[1, x_1, \ldots, x_1^2, x_1x_2, \ldots, x_1x_2x_3, \ldots]$ |
| $\vdots$ | $\vdots$ | $\vdots$ |

# Polynomial Basis

- Let's review XOR problem using a **polynomial basis** as a **feature transformation**
  - **Pick k=2, then we have** $\phi([x_1, x_2]) = [1, x_1, x_2, x_1^2, x_1x_2, x_2^2]$
  - Let's assume we train a perceptron that can finds a separator with coefficients $\theta = [0,0,0,0,4,0]$ and $\theta_0 = 0$
  - This corresponds to $0 + 0x_1 + 0x_2 + 0x_1^2 + 4x_1x_2 + 0x_2^2 + 0 = 0$

# Polynomial Basis

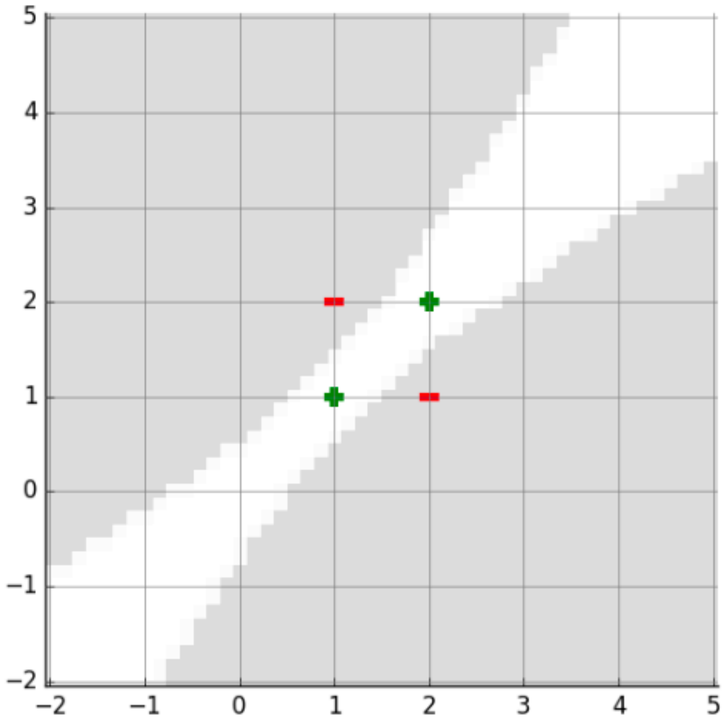
- A separator $0 + 0x_1 + 0x_2 + 0x_1^2 + 4x_1x_2 + 0x_2^2 + 0 = 0$
- Can be plotted as:




- Gray shaded region classified as negative and the white region classified as positive

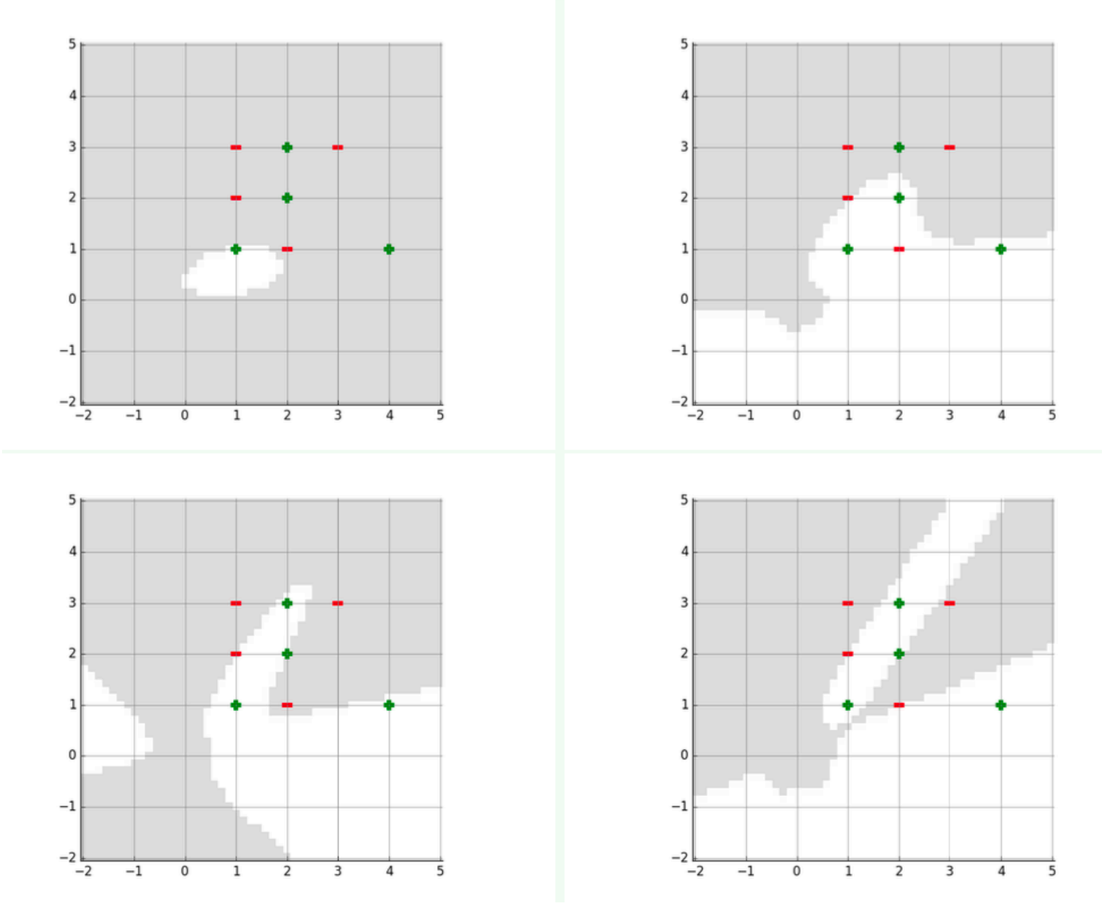
# Other possible separator

- $\theta = [1, -1, -1, -5, 11, -5]$ and $\theta_0 = 1$



# With a Harder Dataset

- After 200 iterations, basis of order 2,3,4, and 5



# Features

- In machine-learning applications, we are given descriptions of inputs with
  - Different type of attributes: numbers, words, discrete features
- Important factor: the way that the features are chosen to be encoded by the human
- Types of features
  - Discrete features
  - Text
  - Numeric values

# Discrete Features

- Assuming the input vectors $x \in \mathbb{R}^d$
- Encoding strategies
  - **Numeric**:
    - When the discrete values really do signify some sort of numeric quantity
    - Strategy: assign each of these values a number $1.0/k,\ 2.0/k, \cdots, 1.0$
  - **Thermometer code**:
    - If your discrete values have a natural ordering, from $1, \cdots, k$, but not a natural mapping into real numbers or numerical quantities.
    - Strategy: vector of length $k$ binary values, where the first $j$ values are 1.0 and the rest are 0.0

# Discrete Features

- Encoding strategies
  - **Factored code**:
    - If your discrete values can be decomposed into several parts (ex, car: ‘year’, and ‘maker’ of the car)
    - Strategy: treat those as two separate features and choose an appropriate encoding of each one from this list
  - **One-hot code**:
    - If there is no obvious metric, ordering, or factorial structure
    - Strategy: use a vector of length $k$, where we convert $0 < j \leq k$ into a vector in which all values are 0.0, except for $j$-th, which is 1.0
  - **Binary code:**
    - $k$ values using vector of length $log(k)$ - bad idea

# Example

- We want to encode blood types, drawn from the set

$$\{A+, A-, B+, B-, AB+, O+, O-\}$$

- **Analysis**:
  - There is no obvious linear numeric scale or even ordering
  - But, there is a reasonable factoring into two features

$$\{A, B, AB, O\} \text{ and } \{+,-\}$$

  - $\{A, B, AB, O\}$ can be even factored like $\{A, \text{not } A\}$, $\{B, \text{not } B\}$

# Example

- We want to encode blood types, drawn from the set

$$\{A+, A-, B+, B-, AB+, O+, O-\}$$

  - Two features: $\{A, B, AB, O\}$ and $\{+,-\}$
  - Note: $\{A, B, AB, O\}$ can be even factored like $\{A, \text{not } A\}, \{B, \text{not } B\}$
- **Strategy**:
  - Use a 6-D vector (4-D for $\{A, B, AB, O\}$) and 2-D for $\{+,-\}$) and apply one-hot encoding for each
  - Use a 3-D vector encoding presence as 1.0 and absence as -1.0. AB+ would be $(1.0, 1.0, 1.0)$ and O- would be $(-1.0, -1.0, -1.0)$

# Other Features

- Texts
  - fixed-length feature vector (will be discussed in sequential models)
  - Bag of words (BoW) model: d be the number words in our vocabulary - 1.0 if that words appear 0.0 or not
- Numeric values
  - Heart rate, stock price, distance, etc: keep it as a numeric value
  - Encoding someone's age
    - Consider under and over 18 (or 21) depending on what kinds of things you are trying to predict
    - One-hot encoding for some sorts of medical situations
  - Useful to **scale** it, such as $[-1, +1]$, or transformation $\phi(x) = \frac{x - \bar{x}}{\sigma}$, where $\bar{x}$ is the average of $x^{(i)}$, and $\sigma$ is the standard deviation of the $x^{(i)}$.

# Other Features

- Numeric values
  - Useful to **scale** it, such as $[-1, +1]$
  - Or transformation like:

$$\phi(x) = \frac{x - \bar{x}}{\sigma},$$

    where $\bar{x}$ is the average of $x^{(i)}$, and $\sigma$ is the standard deviation of the $x^{(i)}$.
  - Or you can apply higher order polynomial-basis transformation

# QnA

- Any questions?
- Note: this lecture notes are largely adopted from
  - MIT 6.036(2019 Fall), Stanford CS229(2023-2024 Summer)